

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA15 16
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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

I **R.JAIDEEP (192525441)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **R.JAIDEEP**

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. **Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.**

Comparison of Type-1 and Type-2 Hypervisors for Mission-Critical Cloud Datacenters

Hypervisors are essential components of cloud datacentre's because they allow multiple virtual machines (VMs) to run on the same physical server while maintaining isolation between workloads. A hypervisor, also called a Virtual Machine Monitor (VMM), manages the allocation of CPU, memory, storage, and network resources to virtual machines. Hypervisors are broadly classified into **Type-1 (bare-metal)** and **Type-2 (hosted)** hypervisors. For cloud datacentre's hosting mission-critical applications, the choice of hypervisor directly affects **performance, security, reliability, scalability, and manageability**.

1. Type-1 Hypervisor

A **Type-1 hypervisor** runs directly on the physical hardware without requiring a conventional host operating system between the hardware and the hypervisor. Examples include **VMware ESXi, Microsoft Hyper-V, and Xen**. Because it directly controls the physical resources, it can efficiently allocate resources among multiple virtual machines.

From a **performance** perspective, Type-1 hypervisors provide low virtualization overhead. CPU, memory, storage, and network operations can be efficiently managed because requests do not need to pass through a general-purpose host operating system. This makes Type-1 hypervisors suitable for applications that require high availability, predictable response times, and continuous processing. Features such as CPU scheduling, memory management, live migration, and resource balancing further improve datacentre performance.

For **manageability**, Type-1 hypervisors are designed specifically for enterprise and datacentre environments. Administrators can centrally manage large numbers of VMs, monitor resource utilization, perform live migration, configure high-availability mechanisms, and automate workload deployment. These capabilities make them highly suitable for large-scale cloud environments.

2. Type-2 Hypervisor

A **Type-2 hypervisor** runs as an application on top of a conventional host operating system such as Windows or Linux. Examples include **Oracle VirtualBox and VMware Workstation**. The architecture is simpler because the host operating system manages the physical hardware while the hypervisor creates and manages virtual machines.

Regarding **performance**, Type-2 hypervisors generally introduce greater overhead than Type-1 hypervisors. VM operations must interact with both the hypervisor and the underlying host operating system. This additional layer can increase latency and reduce resource efficiency, especially when several VMs are running simultaneously. Therefore, Type-2 hypervisors are more appropriate for development, testing, education, and desktop virtualization rather than large production datacenters.

From a **security** perspective, Type-2 hypervisors have a larger attack surface because they depend on the security of the host operating system. If the host operating system becomes compromised, the virtual machines running on it may also be exposed to security risks. Regular updates and strong security controls are therefore necessary. Although Type-2 virtualization provides VM isolation, it is generally less desirable for highly sensitive production workloads.

In terms of **manageability**, Type-2 hypervisors are easy to install and use on individual computers. However, managing hundreds or thousands of VMs across a cloud datacenter becomes more complicated. They lack some of the advanced enterprise features and centralized management capabilities commonly associated with Type-1 virtualization platforms.

3. Comparison		
Factor	Type-1 Hypervisor	Type-2 Hypervisor
Architecture	Runs directly on hardware	Runs on a host OS
Performance	High; low virtualization overhead	Lower due to additional OS layer
Security	Stronger isolation and smaller attack surface	Depends heavily on host OS security
Resource Management	Efficient and centralized	Less efficient for large-scale workloads
Scalability	Excellent for cloud datacenters	Limited for large production environments
Availability	Supports enterprise HA and live migration	Generally limited
Manageability	Advanced centralized management	Easier for individual systems
Typical Usage	Cloud, enterprise, mission-critical applications	Testing, development, desktop virtualization

4. Better Choice for Mission-Critical Cloud Datacenters

For a cloud datacenter hosting **mission-critical applications**, **Type-1 hypervisors are the better choice**. Their direct access to hardware provides better performance and resource utilization, while their reduced dependency on a general-purpose host operating system improves security and reliability. More importantly, enterprise Type-1 platforms support capabilities such as **high availability, live migration, centralized monitoring, resource scheduling, fault tolerance, and automated VM management**.

2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

Workload Isolation in a Cloud Datacenter

A cloud provider may need to host **finance, healthcare, and student portal applications on the same physical hardware** to reduce infrastructure costs and improve resource utilization. However, these applications handle different levels of sensitive information. Finance systems may contain financial and payment information, healthcare systems may store confidential medical records, and student portals may contain academic and personal information. Therefore, strong isolation between workloads is essential. **Virtualization architecture** provides this isolation by allowing multiple independent virtual machines (VMs) to share the same physical server while remaining logically separated.

How Virtualization Architecture Supports Isolation

Virtualization introduces a **hypervisor** between the physical hardware and virtual machines. The hypervisor manages CPU, memory, storage, and network resources and ensures that each VM operates within its allocated resources. For example, the cloud provider can create one VM for the finance application, another VM for the healthcare application, and a third VM for the student portal. Each VM can have its own operating system, applications, virtual disk, virtual network interface, and security policies.

1. Virtual Machine Isolation:

Each workload operates inside a separate VM. The applications running in one VM normally cannot directly access the processes, files, or memory of another VM. If the student portal becomes compromised, virtualization prevents the attacker from automatically accessing the finance or healthcare VM.

2. CPU and Memory Isolation:

The hypervisor allocates virtual CPUs and memory to individual VMs. It maintains separate memory spaces so that one workload cannot directly read another workload's memory. Resource controls can also prevent one VM from consuming all available CPU or memory resources and affecting other critical applications.

3. Storage Isolation:

Each VM can use separate virtual disks or storage volumes. Access-control mechanisms ensure that the finance application can access only its authorized financial data, while the healthcare and student applications have their own storage areas. Encryption can additionally protect sensitive information stored in these virtual disks.

4. Network Isolation:

Virtual networks, VLANs, virtual switches, security groups, and firewalls can be used to separate workloads. For example, the healthcare VM can be placed in a highly restricted network, while the student portal can be placed in a different network segment. Only authorized communication between these environments should be permitted.

5. Access and Security Controls:

Cloud administrators can apply different authentication, authorization, and security policies to each workload. Sensitive finance and healthcare systems can receive stricter access controls than student applications. Monitoring and logging can also detect unusual activities within individual VMs.

Two Major Attack Surfaces

Although virtualization provides strong isolation, there are still important attack surfaces that must be protected.

1. Hypervisor and Virtualization Management Layer:

The hypervisor is one of the most important attack surfaces because it controls all VMs running on the physical server. A vulnerability in the hypervisor could potentially allow an attacker to perform a **VM escape**, in which malicious code running inside one VM attempts to interact with the host or other VMs. Similarly, vulnerabilities in cloud management interfaces or virtualization APIs could allow unauthorized users to modify VM configurations or access resources. Therefore, the hypervisor must be regularly patched, securely configured, monitored, and protected with strong administrative authentication.

2. Virtual Network and Inter-VM Communication:

The virtual networking layer is another major attack surface. Virtual switches, virtual network interfaces, VLAN configurations, firewalls, and security groups control communication between workloads. Incorrect configurations may allow an attacker who compromises the student portal to move laterally toward the finance or healthcare systems. Network attacks such as **spoofing, sniffing, unauthorized access, and lateral movement** can occur when network segmentation is poorly implemented. Strong firewall rules, network segmentation, encryption, and continuous monitoring can reduce these risks.

3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

VM Host Utilization Analysis and Corrective Actions

The given VM host utilization snapshot shows:

- **CPU utilization = 92%**
- **Memory utilization = 88%**
- **Storage I/O latency = 35 ms**
- **Average VM boot time = 170 seconds**

These values indicate that the physical host is under **heavy resource pressure**. The high CPU and memory utilization, combined with relatively high storage latency and slow VM boot time, suggest that the host may be **overcommitted** and unable to provide resources efficiently to all virtual machines.

1. CPU Bottleneck – 92%

A CPU utilization of **92%** is very high for a virtualization host. This indicates that most of the available processing capacity is already being consumed by running VMs and the hypervisor.

The likely causes include:

- Too many VMs running on the same physical host.
- CPU-intensive applications consuming excessive processing power.
- Incorrect allocation of virtual CPUs (vCPUs).
- High CPU contention between VMs.
- Background processes or unnecessary services consuming CPU resources.

2. Memory Bottleneck – 88%

Memory utilization of **88%** also indicates that the host is approaching its physical RAM capacity. High memory usage can cause the hypervisor to use techniques such as memory ballooning, compression, or swapping.

If the host runs out of physical memory, performance can decrease significantly because disk-based swapping is much slower than RAM.

Corrective actions:

- Add more physical RAM to the host.

- Move memory-intensive VMs to another host.
- Identify VMs with excessive or unnecessary memory allocation.
- Optimize applications and reduce unused memory consumption.
- Maintain sufficient memory headroom instead of operating continuously near maximum capacity.

3. Storage I/O Bottleneck – 35 ms Latency

A storage I/O latency of **35 ms** indicates that storage operations are taking a relatively long time. This can significantly affect VM performance because operating systems and applications frequently perform disk reads and writes.

Possible causes include:

- Heavy disk activity from many VMs.
- Slow or overloaded storage devices.
- Insufficient IOPS.
- Large numbers of simultaneous VM operations.
- Storage contention between workloads.

This problem can also explain the **170-second average VM boot time**, because starting a VM requires reading operating-system files, applications, and configuration data from storage.

Corrective actions:

- Move VMs to faster SSD/NVMe-based storage.
- Increase available IOPS and storage throughput.
- Distribute VM disks across multiple storage devices.
- Identify and move storage-intensive VMs.
- Optimize storage configuration and remove unnecessary disk activity.

4. High VM Boot Time – 170 Seconds

An average VM boot time of **170 seconds (almost 3 minutes)** is relatively slow and indicates that the virtualization environment is experiencing resource contention. The most likely contributor is **storage latency**, although CPU and memory pressure can also increase boot time.

During boot, a VM requires CPU processing, memory allocation, and significant storage reads. If the host is already operating at 92% CPU and 88% memory utilization while storage latency is 35 ms, VM startup operations have to compete with existing workloads.

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

Software Defined Datacenter Architecture and Improved Agility

A **Software Defined Datacenter (SDDC)** is a modern datacenter architecture in which compute, storage, and networking resources are virtualized and centrally controlled through software. In a traditional datacenter, most infrastructure components are physically configured and managed separately. This makes deployment slower and requires significant manual effort. SDDC improves **agility, scalability, automation, resource utilization, and management** by allowing infrastructure to be provisioned and modified through software.

1. Compute Virtualization

In a traditional datacenter, applications are usually deployed on dedicated physical servers. If an organization needs additional computing capacity, a new physical server must be purchased, installed, configured, and connected to the network. This process can take considerable time.

SDDC uses **server/compute virtualization** to divide physical servers into multiple virtual machines (VMs). Administrators can create, resize, migrate, or remove VMs through software-based management platforms. Resources such as CPU and memory can be dynamically allocated according to workload requirements.

For example, if an e-commerce application experiences a sudden increase in users, additional VMs can be provisioned quickly. When demand decreases, unnecessary VMs can be removed and resources returned to the resource pool. Technologies such as **live migration, automated provisioning, load balancing, and resource scheduling** further improve flexibility.

Therefore, compute virtualization allows organizations to respond rapidly to changing workloads without continuously purchasing new physical servers.

2. Storage Virtualization

Traditional datacenters often use dedicated storage systems that require manual configuration and allocation. Adding storage capacity or moving data between storage systems can be time-consuming.

In an SDDC, **storage virtualization** combines physical storage resources into logical storage pools. Administrators can allocate storage to applications through software rather than manually configuring individual physical devices. Storage can be provisioned, expanded, migrated, and optimized according to application requirements.

For example, a database application requiring additional storage can receive capacity from an available storage pool without requiring major physical infrastructure changes. Software-defined storage can also support features such as **automated provisioning, replication, snapshots, deduplication, and policy-based storage management**.

This improves agility because storage resources can be delivered quickly and efficiently according to changing business requirements.

3. Network Virtualization

Networking is another major area where SDDC improves agility. In a traditional datacenter, network configuration often requires manually configuring physical switches, routers, firewalls, and VLANs. Changes may require coordination between multiple administrators and can take significant time.

SDDC uses **network virtualization and Software-Defined Networking (SDN)** to separate network control from physical hardware. Virtual networks, switches, routers, firewalls, and security policies can be created and modified through software.

Network virtualization also improves security by allowing policies to be applied to individual workloads rather than relying only on physical network boundaries.

4. Automation and Centralized Management

One of the most important advantages of an SDDC is **centralized, software-based management**. Compute, storage, and network resources can be managed through a common management platform. Infrastructure administrators can use automation and orchestration tools to deploy complete application environments.

For example, a new application may require several VMs, storage volumes, and network configurations. In a traditional datacenter, these components may need to be configured separately. In an SDDC, predefined policies and automated workflows can provision these resources together.

This reduces manual configuration, minimizes human errors, and significantly reduces deployment time.